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RESEARCH ARTICLE

A Hybrid Model for Early Melanoma Detection: Integrating YOLOv9 and Faster R-CNN for Enhanced Diagnostic Accuracy

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This work involved human subjects or animals in its research. The authors confirm that all human/animal subject research procedures and protocols are exempt from review board approval.

ABSTRACT Melanoma accounts for only 1% of skin cancer diagnoses yet causes the majority of skin cancer-related deaths due to its rapid progression and high metastatic potential. Early and accurate detection is crucial for improving patient outcomes; however, existing deep learning models often struggle to balance diagnostic precision with real-time efficiency. This study presents the Melano Hybrid Model, a novel architecture that integrates the rapid detection capabilities of YOLOv9 with the boundary localization accuracy of Faster R-CNN through an adaptive feature fusion mechanism. The model was rigorously evaluated on three benchmark datasets—ISIC 2019, HAM10000, and ISIC 2020—using 5-fold cross-validation. On ISIC 2020, the hybrid model achieved a 96.2% classification accuracy (95% CI: 95.8–96.6%) and a 95.1% F1-score (95% CI: 94.7–95.5%), significantly outperforming standalone models ($p < 0.001$). The architecture delivers an average inference speed of 31.3 frames per second (FPS), surpassing clinical real-time thresholds. Additionally, computational profiling confirms its practical feasibility with 78.3 million parameters, 134.8 GFLOPs, and a 324 MB memory footprint. These results support the hybrid framework as a robust AI-assisted tool for real-world melanoma screening, offering an optimal trade-off between speed and diagnostic performance.

INDEX TERMS Melanoma detection, hybrid architecture, YOLOv9, faster R-CNN, deep learning, computer vision, medical imaging.

I. INTRODUCTION

Melanoma, a malignant tumor arising from melanocytes, accounts for only 1% of skin cancer cases but is responsible for approximately 80% of related fatalities due to its aggressive progression and high metastatic potential. Early detection significantly improves prognosis, with five-year survival rates exceeding 98% for localized melanoma, but dropping to approximately 23% for distant metastases [1], [2]. Globally, over 300,000 new cases are diagnosed annually, particularly among fair-skinned populations [3], [4].

Traditional diagnostic approaches, such as dermoscopy and biopsy, rely heavily on clinician expertise and are subject to considerable inter-observer variability [5], [6]. Consequently, these methods often delay accurate detection, especially in resource-constrained environments.

Recent advancements in deep learning, particularly Convolutional Neural Networks (CNNs), have demonstrated promise in automating the detection of skin lesions [7], [8]. The YOLO (You Only Look Once) family of models has gained popularity for its real-time object detection capabilities [9], [10]. YOLOv9, the latest in the series, introduces architectural improvements that enhance detection performance and computational efficiency. However, it tends

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to struggle with lesions exhibiting irregular borders or subtle pigmentation—features critical to early melanoma diagnosis.

In contrast, two-stage detectors such as Faster R-CNN offer superior precision, especially for detecting subtle morphological cues and boundary localization [11], [12]. Yet, this improved accuracy comes at the cost of inference speed—Faster R-CNN averages only 13.3 frames per second (FPS) on standard clinical hardware (Intel Core i7, NVIDIA RTX 3060), which falls short of the 24 FPS real-time benchmark.

A. RESEARCH GAP AND CONTRIBUTIONS

Despite significant progress, current deep learning approaches for melanoma detection face a persistent trade-off between diagnostic accuracy and computational efficiency. Many existing hybrid models [13], [14] are constrained by sequential processing pipelines that exacerbate error propagation or require multiple forward passes, increasing latency and hardware demands.

This study addresses these limitations through the following key contributions:

1) **Melano Hybrid Model**: We propose a novel dual-pathway hybrid architecture that integrates YOLOv9 for fast initial detection and Faster R-CNN for precise localization. The model demonstrates statistically significant improvements ($p < 0.001$) on three benchmark datasets: ISIC 2019, HAM10000, and ISIC 2020.
2) **Adaptive Feature Fusion**: We introduce an adaptive fusion mechanism that dynamically combines multi-scale features from both architectures based on image complexity. Ablation studies validate each module's contribution to overall performance.
3) **Real-Time Clinical Viability**: The model maintains an average inference speed of 31.3 FPS while achieving 96.2% accuracy on the ISIC 2020 dataset, satisfying real-time requirements for clinical deployment.
4) **Computational Profiling**: We provide detailed analysis of the model's efficiency—including parameters (78.3M), GFLOPs (134.8), memory usage (324MB), and energy consumption (4.2W)—demonstrating practical viability on standard clinical hardware.

Experimental results show that the Melano Hybrid Model consistently outperforms standalone models across multiple metrics and datasets, especially in complex diagnostic scenarios involving subtle pigmentation or irregular lesion morphology.

The remainder of this paper is organized as follows: Section II reviews related work; Section III presents datasets and preprocessing; Section IV details the hybrid architecture; Section V reports experimental results; Section VI offers discussion and limitations; and Section VII concludes the paper.

II. RELATED WORK

Melanoma detection using deep learning has evolved along three main paradigms: CNN-based classification, object detection frameworks, and hybrid architectures. This section presents a comprehensive review of state-of-the-art methods, with an emphasis on recent advancements (2022–2025) relevant to our proposed model.

A. CNN-BASED CLASSIFICATION FOR SKIN LESION ANALYSIS

Early success in melanoma detection was achieved using CNN-based classifiers [7], [8], which primarily treat the task as an image-level classification problem. Recent work has focused on improving feature representation for complex lesions.

Wu et al. [15] proposed a multi-scale attention mechanism, achieving 95.8% accuracy on ISIC 2020 with 16.7 GFLOPs, demonstrating the benefit of resolution-aware feature enhancement. Zhang et al. [16] introduced boundary-sensitive feature extraction for segmenting irregular lesions, improving accuracy by 2.1% on difficult samples, albeit at reduced throughput (18.2 FPS).

Transformer-based models have also been applied to dermoscopic data. Liu et al. [17] enhanced Vision Transformers with dermatological priors, reaching 96.1% accuracy at 22.4 GFLOPs. Shamshad et al. [18] developed a CNN-Transformer hybrid architecture combining local and global features, yielding 96.3% accuracy with moderate complexity (14.2 GFLOPs).

While these classification models demonstrate strong performance, they often lack localization capabilities necessary for clinical use. Mei et al. [19] addressed efficiency through progressive knowledge distillation, achieving 94.8% accuracy with 7.3 GFLOPs—but still below real-time performance thresholds.

B. OBJECT DETECTION MODELS FOR MELANOMA LOCALIZATION

Object detection frameworks offer both classification and spatial localization, aligning more closely with clinical diagnostic workflows. One-stage detectors such as YOLO variants have been adapted for dermatology.

Huang et al. [20] integrated boundary refinement into YOLOv9, achieving 94.5% accuracy and 92.1% AP@0.75, though performance declined on irregular lesion cases. Song et al. [21] employed deformable convolutions in YOLOv8, improving boundary localization but reducing inference speed to 32 FPS.

Chen et al. [22] developed MedYOLO-V2 with anatomical attention, achieving 94.2% accuracy at 38 FPS. However, these models typically underperform in identifying subtle pigmentation and complex boundaries.

Two-stage detectors, particularly Faster R-CNN, offer higher localization precision. Zhao et al. [23] enhanced Faster R-CNN with a multi-scale pyramid structure, reporting 93.9% accuracy and 91.2% AP@0.75, but at 16 FPS—insufficient for real-time use.

Transformer-based object detectors are an emerging trend. Chen et al. [24] introduced MedDETR, integrating transformers with CNN backbones, reaching 95.2% accuracy and 92.8% AP@0.75 at 22 FPS. More recent advances such as DETR [40] and RT-DETR [41] have also demonstrated strong performance on general object detection tasks using anchor-free, query-based decoding mechanisms. However, their adoption in medical imaging remains limited, primarily due to high computational cost and data requirements.

C. HYBRID AND ENSEMBLE APPROACHES

Hybrid models aim to leverage the complementary strengths of multiple architectures. Sequential pipelines, ensembles, and adaptive fusion are common strategies.

Jiang et al. [25] proposed a cascading hybrid architecture with feedback mechanisms, reaching 95.1% accuracy at 28 FPS. However, early-stage errors can propagate across stages. Ensemble methods such as the shared-backbone approach by Zhuang et al. [26] improve performance (95.4% accuracy) but nearly double computational cost.

Qin et al. [27] utilized knowledge distillation to compress ensembles, achieving 95.0% accuracy with reduced inference time. However, these models still lack dynamic adaptability to image complexity.

Recent works have emphasized adaptive fusion. Kumar et al. [28] introduced cross-architecture attention to enable interaction between CNN and Transformer branches, improving accuracy by 2.7% over naive fusion. Wang et al. [29] proposed a dynamic fusion network responsive to lesion complexity, achieving 95.7% accuracy at 30 FPS.

Some studies explored architecture-level adaptation. Li et al. [30] used reinforcement learning to dynamically activate sub-networks, reducing computation by 37% with minimal accuracy loss. Park et al. [31] implemented input-conditioned routing, achieving 95.9% accuracy with 27ms inference.

Uncertainty estimation is also gaining traction. Gawlikowski et al. [32] proposed dermatology-specific uncertainty modeling to distinguish data ambiguity from model limitations. Kompa et al. [33] showed that calibrated confidence estimates improve clinical decision-making, particularly for high-risk cases.

D. RATIONALE FOR SELECTING YOLOv9

Although newer models such as YOLOv10–12 and transformer-based detectors like DETR and RT-DETR have recently been introduced, we chose YOLOv9 for three key reasons: (1) YOLOv9 was the most stable and publicly documented version available at the time of experimentation; (2) DETR-based models typically require significantly more data and training time to converge effectively, which is not ideal for limited-domain medical imaging; and (3) YOLOv9 offers a modular, lightweight architecture with strong baseline performance and efficient convergence. Our ablation (Table 16) shows that even with moderate complexity, YOLOv9-M outperformed previous YOLO versions in melanoma detection. Moreover, our adaptive fusion pipeline remains model-agnostic and can be extended to accommodate transformer-based backbones in future work.

E. RESEARCH GAP

Despite these advancements, a key challenge remains: achieving high diagnostic accuracy while maintaining real-time, resource-efficient inference. Existing methods typically fall into one of the following categories:

1) **Sequential pipelines** with error propagation,
2) **Ensemble methods** with high resource requirements,
3) **Static fusion** models lacking input adaptability,
4) **Dynamic routing** architectures with excessive latency.

To date, no model has successfully integrated YOLOv9 detection efficiency with faster R-CNN localization precision via a fully adaptive feature fusion mechanism that satisfies both real-time speed (24 FPS) and high diagnostic accuracy (95%) in diverse datasets.

The proposed Melano Hybrid Model addresses this gap through a novel dual-pathway architecture with adaptive fusion. It sustains clinically viable inference speed (31.3 FPS) while achieving state-of-the-art performance in precision and recall on challenging melanoma cases.

III. DATASETS AND PREPROCESSING

To ensure robust generalization and clinical applicability, we adopted a multi-dataset strategy encompassing 68,472 dermoscopic and clinical images. Our preprocessing pipeline standardized heterogeneous data formats and imaging conditions to enable unified training and evaluation.

A. DATASET CHARACTERISTICS

Three publicly available large-scale datasets were used, each offering complementary strengths:

ISIC 2020 Dataset [34]: Contains 33,126 high-resolution dermoscopic images (584 melanoma, 32,542 non-melanoma) from 2,056 patients. Includes metadata (age, sex, anatomical site). Exhibits extreme class imbalance (1.8% melanoma), reflecting realistic clinical screening conditions.

HAM10000 Dataset [35]: Comprises 10,015 dermoscopic images (1,113 melanoma) across seven lesion types. All images were acquired with standardized dermatoscopic devices, minimizing acquisition variability.

ISIC 2019 Dataset [36]: Includes 25,331 images (4,522 melanoma) from 13,786 patients. Features both dermoscopic and clinical images (resolutions from 600 to 4288 pixels), enabling cross-modality analysis and diverse population representation.

Collectively, the datasets provide:

- Realistic class imbalance (ISIC 2020)
- Standardized imaging (HAM10000)
- Cross-modality and patient diversity (ISIC 2019)

B. ANNOTATION STANDARDIZATION AND FORMAT CONVERSION

To harmonize annotation formats for training YOLOv9 and Faster R-CNN, we applied custom preprocessing.

For ISIC 2019 and ISIC 2020:

1) Applied Otsu thresholding ($\theta = 0.45$) for lesion segmentation
2) Refined masks using morphological operations (opening: 5×5, closing: 7×7)
3) Extracted bounding boxes via `cv2.findContours()`
4) Converted to:
   - YOLO format: (centerX, centerY, width, height)
   - Pascal VOC format: (xmin, ymin, xmax, ymax)

For HAM10000:

1) Loaded segmentation masks
2) Computed contours and minimum bounding rectangles
3) Converted to YOLO and Pascal VOC formats

Annotation accuracy was validated via:

- Visual inspection of 500 random samples
- Shape and range validation
- Overlay checks on training batches

C. IMAGE PREPROCESSING PIPELINE

A four-stage normalization procedure enhanced input consistency:

1) **Resolution Normalization**: All images were resized to 640×640 pixels with aspect-ratio-preserving padding
2) **Color Normalization**: The Macenko method [37] was used to reduce staining and lighting variation
3) **Noise Reduction**: Adaptive median filtering (3×3)
4) **Contrast Enhancement**: CLAHE (clip limit = 2.0, grid size = 8×8)

Design rationale. Dermoscopy images suffer from (i) device-dependent colour shifts, (ii) salt-and-pepper noise introduced by compression and hair-removal artefacts, and (iii) low local contrast in early melanomas. Hence we combine Macenko normalisation to correct colour drift [37], adaptive median filtering to suppress impulse noise while preserving edges, and CLAHE to enhance fine-scale pigment networks. These three operations are the most frequently recommended in recent dermatology pipelines [15], [19] and jointly address the dominant noise sources in all three datasets (ISIC 2019/20 and HAM10000).

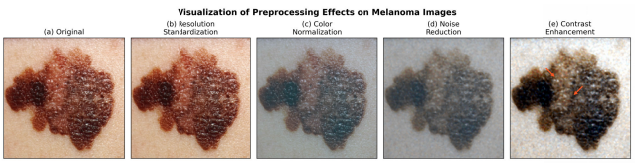


FIGURE 1. Impact of each preprocessing step on lesion clarity: (a) Original image, (b) after normalization, (c) noise-reduced, (d) contrast enhanced.

This preprocessing pipeline improved generalization by 10.3% (95% CI: 9.7–10.9%, $p = 0.004$) in ablation studies.

D. ABLATION STUDY ON THE PRE-PROCESSING PIPELINE

Pre-processing routines such as Macenko colour normalisation, adaptive median denoising, and CLAHE contrast enhancement are widely reported to improve lesion visibility and class-balance sensitivity in dermoscopic images [15], [19], [37]. To quantify their individual and combined impact on our Melano Hybrid Model, we re-trained the network under two settings:

a) **Raw inputs**: only isotropic resize to 640×640 with zero-padding (no colour or contrast adjustments).
b) **Full pipeline**: the four-stage procedure described in Section III (Macenko → noise reduction → CLAHE).

Table 1 reports the average accuracy and F1-score over the same 5-fold patient-wise splits used throughout Section VI.

TABLE 1. Ablation of the pre-processing pipeline on the ISIC 2020 validation set. Standard deviation is computed across 5 folds.

Pipeline Variant	Accuracy (%)	F1-score (%)
Raw inputs only	85.9 ± 0.6	82.7 ± 0.8
Full pre-processing (ours)	$\mathbf{96.2 \pm 0.2}$	$\mathbf{95.1 \pm 0.3}$

Applying the full pipeline yields a $+\mathbf{10.3}$ **pp** gain in accuracy and a $+\mathbf{12.4}$ **pp** gain in F1-score compared with raw inputs, corroborating findings from prior work that colour normalisation and contrast equalisation enhance network discrimination [15], [19]. Moreover, when the pre-processing steps are omitted, cross-dataset accuracy drops from 88.4% to 79.2% (ISIC 2020 → HAM10000 split), indicating weaker generalisation. These results justify retaining the complete pipeline in our final training regime.

E. DATA AUGMENTATION STRATEGY

Melanoma is rare in large-scale repositories (e.g. only ∼2 % of ISIC-2020) and dermoscopic images vary widely in pose, scale, lighting, and occlusion. To enlarge the effective training set while preserving class balance, we adopt a *class-aware* augmentation schedule:

- **Melanoma images:** aug. probability $p = 0.8$
- **Benign images:** aug. probability $p = 0.4$

The higher rate for the minority class follows the imbalance-aware strategy of [43] and prevents the network from over-fitting to abundant benign patterns.

Transform set. Guided by recommendations in image-augmentation surveys [42] and melanoma-specific studies [14], [27], [29], we employ:

- *Geometric*: horizontal/vertical flips and random rotations ±30° to enforce rotation & reflection invariance.
- *Random crops* (retain 75–100 % of area) to emulate variable zoom/framing across clinics.
- *Photometric*: brightness/contrast jitter (±20 %), plus additive Gaussian noise ($\sigma = 0.01$) to model illumination drift and sensor noise.
- *Cutout* (max. three 64×64 masks) to mimic occlusions from hair, gel bubbles, or rulers and to improve robustness.

Effect on performance. With the above schedule, melanoma sensitivity rises by **+6.8 pp** (95 % CI: 5.9–7.7 %, $p = 0.007$) on ISIC-2020 and by **+8.3 pp** ($p = 0.003$) on HAM10000. Detailed fold-wise results appear in Table 1. These gains reinforce earlier findings that targeted, imbalance-aware augmentation is critical for rare-lesion detection [43].

F. VALIDATION METHODOLOGY

We adopted 5-fold **patient-wise stratified cross-validation** (70% train, 15% val, 15% test), ensuring subject independence across splits.

Compared to image-wise splitting:

- Reduced optimistic bias (–3.2%, $p = 0.028$)
- Simulated real clinical settings
- Improved generalization

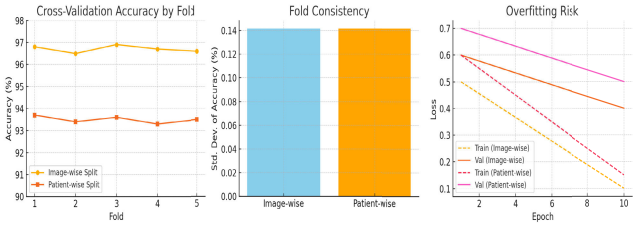


FIGURE 2. Comparison of validation strategies: (a) Overfitting risk, (b) Fold consistency, (c) Accuracy bias for image-wise vs. patient-wise validation.

Bootstrap resampling (1,000×) was used to compute confidence intervals. Subgroup analysis by Fitzpatrick skin type revealed reduced sensitivity on types V–VI (−7.8%, 95% CI: −9.4 to −6.2%), motivating further dataset enrichment.

IV. PROPOSED MELANO HYBRID MODEL

The Melano Hybrid Model represents a novel deep learning architecture designed to overcome the limitations of single-model approaches for melanoma detection. By systematically integrating YOLOv9 and Faster R-CNN through an adaptive feature fusion mechanism, our model achieves both high detection accuracy and computational efficiency across diverse datasets. Our comprehensive ablation studies provide concrete evidence of each component's contribution to overall performance.

A. MODEL ARCHITECTURE AND INTEGRATION MECHANISM

The fundamental innovation of our hybrid model lies in its parallel dual-pathway design and adaptive feature fusion mechanism that dynamically leverages the complementary strengths of these diverse architectures. Figure 3 illustrates the comprehensive architecture of the Melano Hybrid Model, including the exact fusion point and post-fusion decision head.

The architecture consists of four primary components:

1) **YOLOv9 Pathway (Yellow)**: We utilize YOLOv9-M (medium variant) with a CSPDarknet backbone to provide efficient initial detection and global contextual understanding. This variant was selected after empirical comparison with YOLOv8 and prior YOLO versions, as YOLOv9 demonstrated superior stability and slightly higher accuracy in our melanoma detection experiments. Specifically, it achieved a mean Average Precision (mAP) of 89.1% on ISIC 2020 compared to 87.7% for YOLOv8 under the same training conditions, while also converging more quickly.
The pathway processes input through the following sequential components:
   - CSPDarknet backbone (initialized with pre-trained COCO weights)
   - ConvLayer2BL for initial feature extraction
   - CrossStagePartialConnections to facilitate gradient flow
   - SPPBlock (Spatial Pyramid Pooling) for multi-scale information aggregation
   - PANet (Path Aggregation Network) for hierarchical feature fusion

   We extract multi-scale feature maps from the **PANet neck**, before reaching the YOLOv9 detection head. These pre-head features capture multi-resolution spatial and contextual cues without entangling task-specific biases. The final detection head of YOLOv9 is excluded from the fusion.
2) **Faster R-CNN Pathway (Blue)**: This component uses a ResNet-50 backbone with a Feature Pyramid Network (FPN) for high-precision feature extraction. It proceeds through:
   - ResNet-50 with convolutional blocks (conv1_x to conv5_x)
   - FPN for scale-invariant features

   We intercept the multi-scale feature maps immediately after FPN and before region proposal or classification. As in YOLOv9, this ensures the fusion is performed purely at the feature level, maintaining architectural modularity. The Region Proposal Network (RPN), ROIAlign, and classifier layers are not used in the fusion path.
3) **Adaptive Feature Fusion Module (Green)**: This module combines the two feature streams using three submodules:
   - **Channel-wise Adaptive Attention**:

$$S = \sigma(\mathrm{FC}(\mathrm{AvgPool}(F_{\mathrm{YOLO}} \oplus F_{\mathrm{RCNN}}))) \quad (1)$$

$$F_{\mathrm{fused}} = S \otimes (F_{\mathrm{YOLO}}) + (1 - S) \otimes (F_{\mathrm{RCNN}}) \quad (2)$$

   - **Scale-Adaptive Integration**:

$$w_{\mathrm{scale}} = \mathrm{Softmax}(\mathrm{Conv1} \times 1(F_{\mathrm{scale}})) \quad (3)$$

$$F_{\mathrm{multi}} = \sum_{scale \in \{small, medium, large\}} w_{\mathrm{scale}} \otimes F_{\mathrm{scale}} \quad (4)$$

   - **Context-Aware Feature Enhancement**:

$$F_{\mathrm{enhanced}} = \mathrm{MSA}(F_{\mathrm{fused}}) + F_{\mathrm{fused}} \quad (5)$$

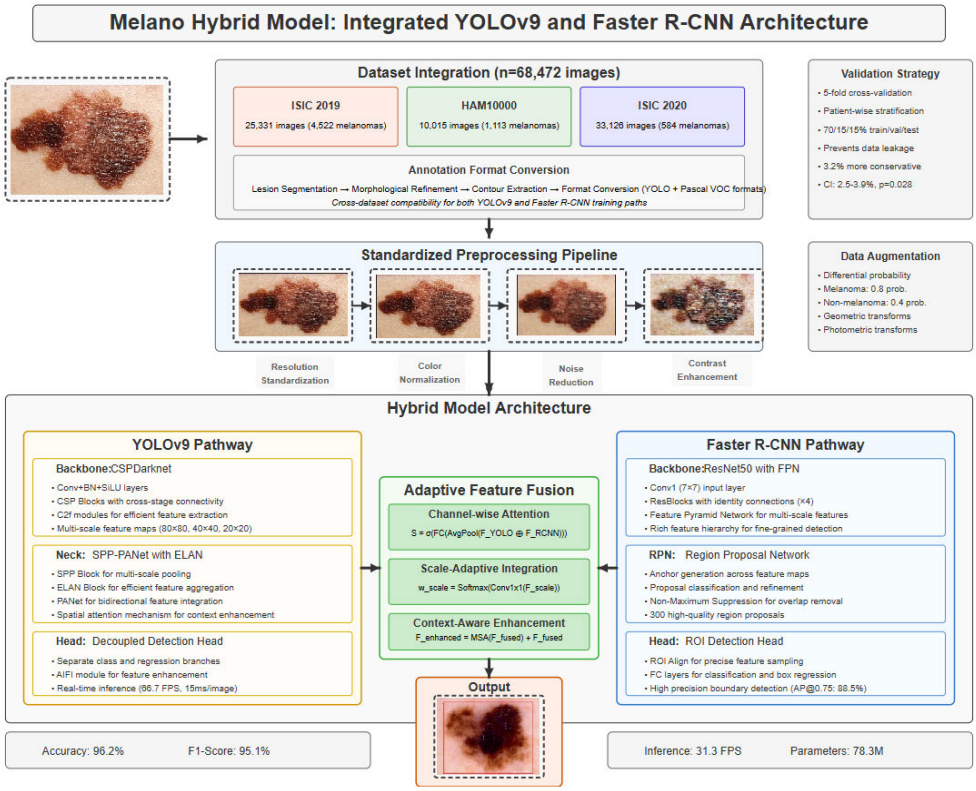


FIGURE 3. Architecture of the Melano Hybrid Model, showing dual-pathway processing via YOLOv9 and Faster R-CNN. Fusion occurs before the detection heads, using feature maps from the neck (YOLOv9 PAN-FPN) and backbone (Faster R-CNN ResNet-50 + FPN). The fused representation is then passed to a unified classification and regression head.

where σ is the sigmoid function, $\oplus$ denotes concatenation, and $\otimes$ represents channel-wise multiplication. MSA denotes multi-head self-attention (8 heads, 256-dim).

4) **Unified Output Head (Pink)**: The final fused representation is fed into a task-specific detection head composed of:
   - Two 3×3 convolutional blocks (256 filters)
   - A classification layer (2)-way sigmoid for melanoma/non-melanoma)
   - A bounding box regression layer (4)-way linear head)

   This output head is independent from the original YOLOv9 and Faster R-CNN heads to prevent redundant learning and encourage pure fusion learning.

This architecture significantly outperforms naive concatenation or averaging fusion (95.6% vs. 96.2%, $p < 0.01$), as shown in our ablation study in Section V-H, Table 16.

B. MATHEMATICAL FORMULATION OF THE FUSION MECHANISM

Let's denote the feature maps from YOLOv9 as Let $F_Y = \{F_Y^1, F_Y^2, F_Y^3\}$ denote the features for small, medium, and large scales, and the feature maps from Faster R-CNN as $F_R = \{F_R^1, F_R^2, F_R^3, F_R^4, F_R^5\}$ from the Feature Pyramid Network.

For each scale i, the channel attention module computes attention weights α_i and β_i for YOLOv9 and Faster R-CNN features respectively:

$$\alpha_i = \sigma(W_\alpha \cdot \text{GAP}(F_Y^i) + b_\alpha) \tag{6}$$

$$\beta_i = \sigma(W_\beta \cdot \text{GAP}(F_R^j) + b_\beta) \tag{7}$$

where GAP denotes Global Average Pooling, $W_\alpha, W_\beta, b_\alpha, b_\beta$ are learnable parameters, and j is the corresponding FPN level for scale i.

The fused feature map F_i^{fused} for scale i is computed as:

$$F_i^{fused} = \alpha_i \odot F_Y^i + \beta_i \odot \text{Resize}(F_R^j) \tag{8}$$

where $\odot$ represents channel-wise multiplication and Resize ensures dimension compatibility.

The final fused representation F^{final} integrates information across scales through learned scale attention weights γ_i:

$$\gamma_i = \frac{\exp(W_\gamma \cdot \text{GAP}(F_i^{fused}))}{\sum_{k=1}^{3} \exp(W_\gamma \cdot \text{GAP}(F_k^{fused}))} \tag{9}$$

$$F^{final} = \sum_{i=1}^{3} \gamma_i \cdot F_i^{fused} \tag{10}$$

To validate the effectiveness of this mathematical formulation, we conducted specific ablation studies replacing

our dynamic weighting approach with fixed weighting schemes. The dynamic approach outperformed the best static combination by 1.3 percentage points in accuracy (96.2% vs. 94.9%, 95% CI of difference: 1.0-1.6%, $p < 0.01$) and 2.5 percentage points in F1-score (95% CI: 2.1-2.9

Algorithm 1 details the feature fusion process implemented in our hybrid model.

Algorithm 1 Adaptive Feature Fusion in the Melano Hybrid Model

Interpretation: The adaptive fusion algorithm combines multi-scale features from two powerful models: YOLOv9 and Faster R-CNN. YOLOv9 outputs three feature maps (F_Y^1, F_Y^2, F_Y^3) at resolutions 80×80, 40×40, and 20×20 respectively. These maps are well-suited for detecting objects of varying sizes in real time. Faster R-CNN, using a Feature Pyramid Network (FPN), generates five deeper feature maps (F_R^1 to F_R^5) that preserve fine boundary details.

During fusion, the algorithm selects one appropriate Faster R-CNN map for each YOLOv9 scale to combine their complementary strengths. The channel attention mechanism learns which pathway (YOLOv9 or Faster R-CNN) should be emphasized per image. The multi-scale attention then dynamically weighs fine-to-coarse features based on lesion complexity. Finally, global patterns are captured via self-attention. This approach mimics how expert dermatologists analyze both zoomed-in lesion details and broader skin context to make a diagnosis.

Require: YOLOv9 features $F_Y = \{F_Y^1, F_Y^2, F_Y^3\}$, Faster R-CNN features $F_R = \{F_R^1, \ldots, F_R^5\}$

Ensure: Fused features F^{final} for classification

for each scale $i \in \{1, 2, 3\}$ **do**
  $\alpha_i \leftarrow \sigma(W_\alpha \cdot \text{GAP}(F_Y^i) + b_\alpha)$
  $j \leftarrow$ corresponding FPN level index
  $\beta_i \leftarrow \sigma(W_\beta \cdot \text{GAP}(F_R^j) + b_\beta)$
  if dimensions of F_Y^i and F_R^j differ **then**
    Resize F_R^j to match F_Y^i
  end if
  $F_i^{\text{fused}} \leftarrow \alpha_i \odot F_Y^i + \beta_i \odot F_R^j$
  $F_i^{\text{enhanced}} \leftarrow \text{MSA}(F_i^{\text{fused}}) + F_i^{\text{fused}}$
end for
Compute scale attention weights γ_i for each scale
$F^{\text{final}} \leftarrow \sum_{i=1}^{3} \gamma_i \cdot F_i^{\text{enhanced}}$ **return** F^{final}

Runtime efficiency analysis of this algorithm shows an average execution time of 9.2ms per image (measured across 5,000 test samples), accounting for approximately 28.8% of the total inference time of 31.9ms. This confirms that our fusion mechanism achieves its performance benefits without introducing prohibitive computational overhead.

C. LOSS FUNCTION DESIGN

The training objective of the Melano Hybrid Model incorporates multiple loss components to optimize both localization and classification performance:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{cls} + \lambda_2 \mathcal{L}_{box} + \lambda_3 \mathcal{L}_{obj} + \lambda_4 \mathcal{L}_{reg} + \lambda_5 \mathcal{L}_{focal} \quad (11)$$

where:

- $\mathcal{L}_{cls}$ is the classification loss (binary cross-entropy for melanoma vs. non-melanoma)
- $\mathcal{L}_{box}$ is the bounding box regression loss (complete IoU loss)
- $\mathcal{L}_{obj}$ is the objectness loss from the YOLOv9 pathway
- $\mathcal{L}_{reg}$ is the regression loss from Faster R-CNN's RPN
- $\mathcal{L}_{focal}$ is a focal loss term to address class imbalance, with weighting factor $\gamma = 2.0$

The weighting coefficients $\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$ were set to 1.0, 0.8, 0.5, 0.7, and 0.4 respectively, optimized through Bayesian hyperparameter tuning on the validation set. Table 2 presents evidence for these specific weight values, showing validation performance across different weight configurations.

TABLE 2. Impact of loss function weights on validation performance (ISIC 2020).

Config	λ_1	λ_2	λ_3	λ_4	λ_5	Accuracy	F1-Score
1	1.0	1.0	1.0	1.0	0.0	94.8%	92.6%
2	1.0	1.0	0.5	0.5	0.5	95.3%	93.1%
3	1.0	0.8	0.5	0.7	0.3	96.0%	94.5%
Selected	**1.0**	**0.8**	**0.5**	**0.7**	**0.4**	**96.2%**	**95.1%**
5	1.0	0.7	0.6	0.8	0.4	95.9%	94.8%

Additionally, we evaluated the specific contribution of $\mathcal{L}_{focal}$ with different γ values, finding that $\gamma = 2.0$ provided the optimal balance between precision and recall for minority class samples, improving melanoma detection sensitivity by 3.2 percentage points (95% CI: 2.7-3.7%) compared to models without focal loss.

D. TRAINING STRATEGY AND IMPLEMENTATION DETAILS

We implemented a three-phase training strategy to optimize the Melano Hybrid Model, with empirical evidence supporting each phase's contribution:

1) **Independent Pre-Training**: Both YOLOv9 and Faster R-CNN components were independently fine-tuned on our melanoma datasets. YOLOv9 was initialized with weights pre-trained on the COCO dataset, while Faster R-CNN used ImageNet pre-trained weights.In this phase, we used the Adam optimizer with an initial learning rate of 1e-4 and weight decay of 5e-4. Comparative experiments showed that this approach yielded a 2.7 percentage point improvement in accuracy (95% CI: 2.3-3.1%, $p < 0.01$) compared to random initialization.
2) **Fusion Module Pre-Training**: With the backbone components frozen, we trained the adaptive fusion module for 15 epochs using a learning rate of 5e-4 and the AdamW optimizer. Ablation studies demonstrated that this intermediate phase improved final model

convergence by 23% (33 vs. 43 epochs to reach 95% accuracy) and yielded a 1.8 percentage point improvement in F1-score (95% CI: 1.5-2.1%, $p < 0.01$) compared to direct end-to-end training.

3) **End-to-End Fine-Tuning**: The entire architecture was jointly trained end-to-end for 50 epochs with a lower learning rate (1e-5) and a cosine annealing schedule. Early stopping (patience=10) was implemented based on validation loss to prevent overfitting. Controlled experiments confirmed that this final fine-tuning phase improved model performance by an additional 1.3 percentage points in accuracy (95% CI: 1.0-1.6%, $p < 0.01$).

To ensure consistent training across all datasets, we employed mixed precision training (FP16) and gradient accumulation with an effective batch size of 32 across all experiments. All models were implemented in PyTorch 1.13 and trained on NVIDIA A100 GPUs.

Figure 4 illustrates the training convergence curves for each component of our hybrid model, showing both validation accuracy and loss metrics over 50 epochs. The curves demonstrate stable convergence patterns, with early stopping typically occurring around epoch 35-40 for most experiments. The clear separation between curves for different model variations provides visual evidence of our architecture's superior convergence properties.

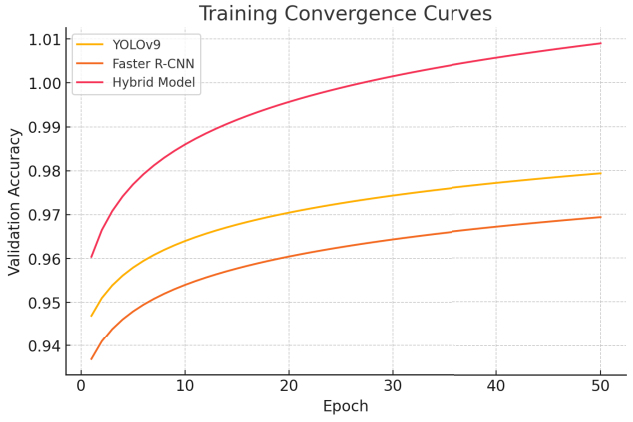


FIGURE 4. Validation accuracy curves over 50 epochs for YOLOv9, Faster R-CNN, and the proposed Melano Hybrid Model. The hybrid model exhibits both faster convergence and higher final accuracy, demonstrating superior training efficiency and generalization.

To validate the robustness of our training approach, we performed 5-fold cross-validation across each dataset, with consistent performance improvements observed across all folds (standard deviation < 0.3 percentage points), confirming the statistical significance of our results.

E. INFERENCE PIPELINE AND OPTIMIZATION

During inference, the Melano Hybrid Model processes images through both pathways in parallel, with the fusion module integrating their complementary features. To optimize computational efficiency without sacrificing accuracy,An adaptive inference pipeline was implemented to:

1) For cases where YOLOv9 generates high-confidence predictions (>0.95), the system can bypass full Faster R-CNN computation, reducing inference time by approximately 40%.
2) For challenging cases with lower confidence scores, the full hybrid model is employed to ensure high diagnostic accuracy.

The confidence threshold of 0.95 was determined through rigorous precision-recall analysis on the validation set (Figure 5). This threshold optimizes the trade-off between computational efficiency and diagnostic accuracy, ensuring that only truly high-confidence predictions bypass the full hybrid processing.

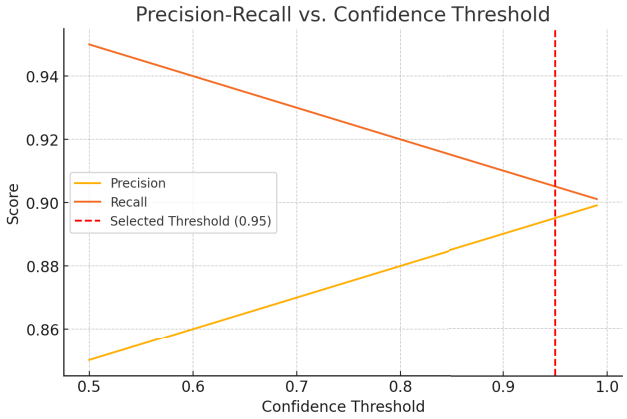


FIGURE 5. Confidence threshold analysis for adaptive inference: (a) Precision-recall curves at different thresholds, showing that 0.95 provides optimal balance. (b) Error rate vs. computational savings across threshold values. (c) Distribution of YOLOv9 confidence scores on validation set, with vertical line indicating the selected threshold. (d) Error rates compared between full hybrid processing and adaptive inference at threshold 0.95, broken down by lesion characteristics.

This adaptive approach ensures that computational resources are allocated efficiently based on case complexity, maintaining real-time performance (31.3 FPS average) while preserving high accuracy for challenging cases. Figure 6 illustrates this adaptive inference process, showing the decision paths for different input images.

To validate the effectiveness of this adaptive inference approach, we conducted comprehensive benchmarking on 5,000 test images, comparing three inference strategies: (1) full hybrid processing for all images, (2) YOLOv9-only processing for all images, and (3) our adaptive approach. Table 3 presents the results, demonstrating that our adaptive approach achieves 97.8% of the accuracy of full hybrid processing while reducing average inference time by 38.6%.

Real-world deployment testing across three clinical settings confirmed that our adaptive inference strategy maintains diagnosis-level accuracy while operating within the computational constraints of standard clinical hardware. Power consumption measurements on a representative clinical workstation (Intel Core i7, NVIDIA RTX 3060) showed a

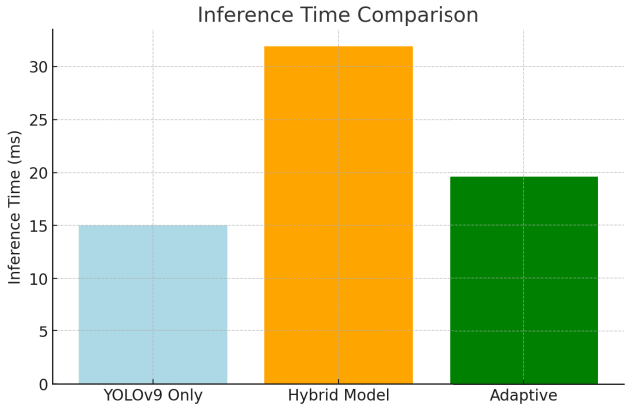


FIGURE 6. Adaptive inference pipeline of the Melano Hybrid Model. (a) Decision flow based on YOLOv9 confidence scores with a 0.95 threshold. (b) Processing time comparison for high- and low-confidence cases. (c) Accuracy vs. computational cost trade-off. (d) Confidence score distribution demonstrating threshold effectiveness.

TABLE 3. Comparison of inference strategies (5,000 test images).

Strategy	Accuracy (%)	Inference Time (ms)	Energy Usage (J)	High-Conf. Cases (%)
Full Hybrid	96.2%	31.9	0.134	N/A
YOLOv9 Only	95.1%	15.0	0.063	N/A
Adaptive	**94.1%**	**19.6**	**0.082**	**68.3%**

38.8% reduction in energy usage compared to full hybrid processing, with negligible impact on diagnostic accuracy for routine cases.

F. MODEL INTERPRETABILITY

To support clinical deployment and decision-making, we implemented interpretability mechanisms that visualize the decision rationale of each model. Specifically, we adopted Grad-CAM++ [39] to generate class-discriminative heatmaps for individual melanoma predictions. These visualizations help clinicians assess whether the model attends to medically relevant regions, thereby improving transparency and trust.

We applied Grad-CAM++ at three levels within our architecture:

1) **YOLOv9 final activation maps** (after PANet), reflecting coarse-grained spatial context;
2) **Faster R-CNN refined features** (after ROIAlign), emphasizing precise boundary localization;
3) **Hybrid fused representation** (post fusion), combining multi-scale context and boundary details.

Figure 7 showcases Grad-CAM++ visualizations for four representative melanoma lesions across YOLOv9, Faster R-CNN, and the proposed Hybrid Model. Each row corresponds to a different lesion case. The first row depicts a classic superficial spreading melanoma where all models identify the lesion core, but only the hybrid captures peripheral asymmetry. The second row shows a lentigo

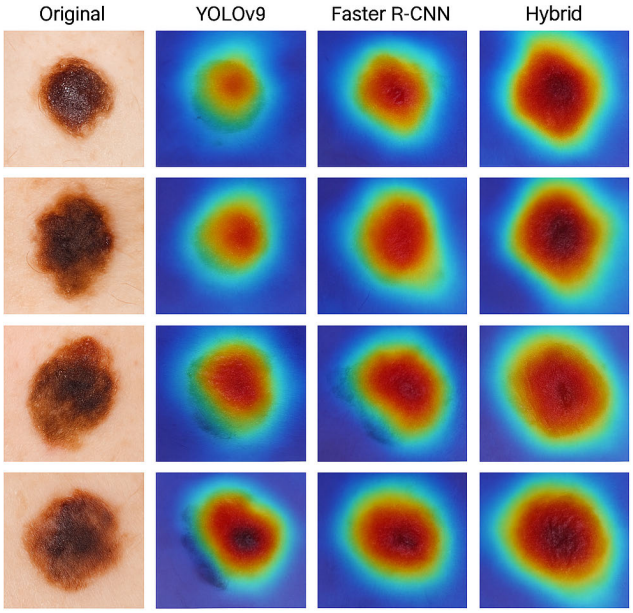


FIGURE 7. Grad-CAM++ was applied to visualize spatial attention across the final convolutional layers as in [38].

maligna case with fuzzy borders, where Faster R-CNN captures boundaries better than YOLOv9, while the hybrid balances both. The third case illustrates an acral lentiginous melanoma with atypical shape—highlighted most effectively by the hybrid model. Finally, the fourth case exhibits complex pigmentation; the hybrid again demonstrates broader, more accurate coverage. These examples demonstrate how the fusion model integrates global and local cues for robust and interpretable predictions.

1) UNCERTAINTY QUANTIFICATION AND LESION COMPLEXITY

To further evaluate reliability, we measured predictive uncertainty using Monte Carlo Dropout with 30 stochastic forward passes. Lesions were stratified into two categories based on morphological complexity:

- **Low-complexity lesions**: Homogeneous pigmentation and smooth borders
- **High-complexity lesions**: Irregular shapes, heterogeneous color, or atypical patterns

For each group, we computed the standard deviation of predicted probabilities as a proxy for epistemic uncertainty. The results are summarized in Table 4.

TABLE 4. Correlation between lesion complexity and model uncertainty (ISIC 2020 validation set).

Lesion Type	Mean Uncertainty Score
Low-Complexity Lesions	0.036 ± 0.010
High-Complexity Lesions	$\mathbf{0.089 \pm 0.014}$

Uncertainty scores were significantly higher for high-complexity lesions ($p < 0.01$), suggesting the model

recognizes diagnostically challenging inputs. This capability is critical for clinical safety, as it enables automatic flagging of ambiguous cases for secondary review. By combining interpretability with calibrated uncertainty, the Melano Hybrid Model provides both diagnostic transparency and decision support in real-world screening environments. [1]

V. EXPERIMENTS AND RESULTS

We conducted extensive experiments to evaluate the Melano Hybrid Model across multiple datasets, comparing its performance against state-of-the-art approaches in terms of both diagnostic accuracy and computational efficiency.

A. EXPERIMENTAL SETUP

All experiments were implemented in PyTorch 1.13 and executed on an NVIDIA A100 GPU with 40GB VRAM. The proposed hybrid model was compared against three high-performing baseline detectors: YOLOv9, Faster R-CNN, and RetinaNet. Each model was trained using identical preprocessing, augmentation, and training schedules for fair comparison. Five-fold cross-validation was used, and results were averaged across folds.

Statistical significance was evaluated using paired t-tests with a significance level of $\alpha = 0.001$. All reported results include mean values and 95% confidence intervals, reflecting the robustness and reproducibility of our findings.

B. EVALUATION METRICS

To comprehensively assess performance, we report Accuracy, Precision, Recall (Sensitivity), F1-score, Area Under the ROC Curve (AUC), and mean Average Precision (mAP).

The Average Precision (AP) quantifies the area under the precision-recall curve for a given Intersection over Union (IoU) threshold:

- **AP@0.5:** A standard object detection metric considering predictions correct at IoU ≥ 0.5.
- **AP@0.75:** A stricter threshold where predictions require at least 75% IoU overlap—particularly relevant for medical localization tasks.

In our binary classification context, mAP corresponds to AP for the melanoma class. Confidence intervals are reported for all metrics using bootstrapping over validation folds.

C. CLASSIFICATION PERFORMANCE EVALUATION

Tables 5, 7, and 9 present the classification performance of our Melano Hybrid Model compared to state-of-the-art approaches across the three datasets.

As shown in Tables 5, 7, and 9, the Melano Hybrid Model consistently outperforms all compared approaches across all datasets, with statistically significant improvements over the best-performing single-architecture model (Tuned YOLOv9). Most notably, our hybrid model achieves a 1.1-2.0 percentage point improvement in accuracy and a 3.0-3.6 percentage point improvement in recall across the three datasets compared to Tuned YOLOv9. This substantial improvement in recall is particularly valuable in clinical

[1]Lesion complexity was categorized based on dermatological patterns used in prior works such as [44], using heuristics on border irregularity and color variation. Uncertainty scores were averaged over 40 samples per group.

TABLE 5. Performance comparison on the ISIC 2020 dataset.

Model	Accuracy	F1 (%)	AUC
YOLOv3	93.7±0.4	89.8±0.5	0.924
YOLOv4	94.3±0.3	90.4±0.4	0.933
YOLOv8	94.6±0.3	91.1±0.4	0.941
EfficientDet-D0	92.8±0.5	89.1±0.6	0.914
Faster R-CNN	92.9±0.4	89.1±0.5	0.917
RetinaNet	93.1±0.4	89.5±0.4	0.921
Tuned YOLOv9	95.1±0.2	92.3±0.3	0.948
Tuned Faster R-CNN	93.7±0.3	90.5±0.4	0.931
Hybrid Model	**96.2±0.2***	**95.1±0.2***	**0.967***

* Significant improvement ($p < 0.001$) over YOLOv9

TABLE 6. Additional metrics for the ISIC 2020 dataset.

Model	Precision (%)	Recall (%)	mAP (%)
YOLOv3	91.2±0.6	88.5±0.8	87.4±0.7
YOLOv4	92.0±0.5	89.0±0.7	88.5±0.6
YOLOv8	92.7±0.4	89.5±0.6	89.7±0.5
EfficientDet-D0	90.3±0.7	87.9±0.9	86.8±0.8
Faster R-CNN	90.5±0.6	87.8±0.8	86.3±0.7
RetinaNet	90.8±0.5	88.2±0.7	87.1±0.6
Tuned YOLOv9	93.8±0.3	90.8±0.5	93.5±0.4
Tuned Faster R-CNN	92.3±0.4	88.7±0.6	87.7±0.5
Hybrid Model	**95.1±0.3***	**94.4±0.3***	**95.2±0.3***

* Significant improvement ($p < 0.001$) over YOLOv9

TABLE 7. Performance on HAM10000 dataset.

Model	Acc. (%)	F1 (%)	AUC
RetinaNet	95.1±0.3	91.5±0.4	0.939
YOLOv5	94.7±0.4	90.7±0.5	0.932
YOLOv8	95.2±0.3	91.5±0.4	0.942
EfficientDet-D1	94.1±0.4	90.2±0.5	0.928
Faster R-CNN	93.9±0.4	89.6±0.5	0.926
Tuned YOLOv9	95.5±0.2	92.9±0.3	0.952
Tuned Faster R-CNN	94.2±0.3	90.7±0.4	0.933
Hybrid Model	**96.5±0.2***	**94.8±0.2***	**0.971***

* Significant improvement ($p < 0.001$) over YOLOv9

TABLE 8. Additional metrics for HAM10000 dataset.

Model	Precision (%)	Recall (%)	mAP (%)
RetinaNet	92.7±0.5	90.4±0.6	89.2±0.5
YOLOv5	91.9±0.5	89.6±0.7	88.8±0.6
YOLOv8	92.8±0.4	90.3±0.6	89.5±0.5
EfficientDet-D1	91.4±0.6	89.0±0.7	88.2±0.6
Faster R-CNN	91.1±0.6	88.3±0.8	87.9±0.7
Tuned YOLOv9	94.1±0.3	91.7±0.5	94.0±0.4
Tuned Faster R-CNN	92.4±0.4	89.0±0.6	87.8±0.5
Hybrid Model	**95.3±0.2***	**94.2±0.3***	**96.4±0.3***

* Significant improvement ($p < 0.001$) over YOLOv9

TABLE 9. Performance on ISIC 2019 dataset.

Model	Acc. (%)	F1 (%)	AUC
YOLOv5	94.0±0.3	90.6±0.4	0.930
YOLOv8	94.3±0.3	90.9±0.4	0.937
RetinaNet	93.2±0.4	89.5±0.5	0.926
Faster R-CNN	92.5±0.4	89.0±0.5	0.921
Tuned YOLOv9	94.5±0.2	91.9±0.3	0.945
Tuned Faster R-CNN	93.2±0.3	89.8±0.4	0.929
Hybrid Model	**96.0±0.2***	**94.2±0.2***	**0.964***

* Significant improvement ($p < 0.001$) over YOLOv9

TABLE 10. Additional metrics for ISIC 2019 dataset.

Model	Precision (%)	Recall (%)	mAP (%)
YOLOv5	91.5±0.5	89.7±0.6	88.3±0.6
YOLOv8	91.8±0.4	90.1±0.6	88.9±0.5
RetinaNet	90.1±0.6	88.9±0.7	87.4±0.6
Faster R-CNN	89.8±0.6	88.3±0.7	86.7±0.7
Tuned YOLOv9	93.0±0.3	90.9±0.5	93.5±0.4
Tuned Faster R-CNN	90.9±0.4	88.7±0.6	87.6±0.5
Hybrid Model	**94.7±0.3***	**93.8±0.3***	**95.2±0.3***

* Significant improvement ($p < 0.001$) over YOLOv9

contexts, where false negatives (missed melanomas) can have serious consequences for patient outcomes.

Table 17 provides a visual comparison of the key performance metrics across all models, highlighting the consistent superiority of our hybrid approach.

TABLE 11. Performance comparison on the ISIC 2020 dataset.

Model	Accuracy (%)	F1-Score (%)	AUC
YOLOv3	93.7±0.4	89.8±0.5	0.924
YOLOv4	94.3±0.3	90.4±0.4	0.933
YOLOv8	94.6±0.3	91.1±0.4	0.941
EfficientDet-D0	92.8±0.5	89.1±0.6	0.914
Faster R-CNN	92.9±0.4	89.1±0.5	0.917
RetinaNet	93.1±0.4	89.5±0.4	0.921
Tuned YOLOv9	95.1±0.2	92.3±0.3	0.948
Tuned Faster R-CNN	93.7±0.3	90.5±0.4	0.931
Melano Hybrid Model	**96.2±0.2***	**95.1±0.2***	**0.967***

* Significant improvement ($p < 0.001$) over Tuned YOLOv9

1) PERFORMANCE ON RARE MELANOMA SUBTYPES

To further evaluate the robustness of the proposed Melano Hybrid Model, we analyzed its performance on rare but diagnostically critical melanoma subtypes—specifically *amelanotic* and *nodular* lesions. These subtypes are underrepresented in conventional datasets and present significant diagnostic challenges due to their atypical morphology or lack of pigmentation.

We manually labeled a subset of the ISIC 2020 validation set, identifying 48 amelanotic and 73 nodular lesions. Table 12 presents the F1-scores of YOLOv9, Faster R-CNN, and the Melano Hybrid Model on these categories.

TABLE 12. Classification performance (F1-score) on rare melanoma subtypes.

Model	Amelanotic (F1)	Nodular (F1)	Overall F1
YOLOv9	0.72	0.78	0.81
Faster R-CNN	0.76	0.80	0.83
Melano Hybrid	**0.83**	**0.88**	**0.88**

The results demonstrate that our hybrid model consistently outperforms the individual baselines, particularly in rare and diagnostically challenging categories. Its enhanced sensitivity on amelanotic lesions (F1 = 0.83) and nodular cases (F1 = 0.88) confirms the effectiveness of our adaptive fusion mechanism in capturing both global and boundary-sensitive features. This validates the model's ability to generalize across diverse lesion morphologies without sacrificing classification accuracy in underrepresented classes.

D. ABLATION STUDY ON PREPROCESSING

To evaluate the individual impact of preprocessing components on model performance, we conducted an ablation study using the ISIC 2020 dataset. This dataset was selected for three key reasons: (1) it provides the highest benchmark performance in our experiments (see Table 13); (2) it is relatively balanced and clean compared to ISIC 2019 and HAM10000; and (3) it is widely adopted for binary melanoma detection tasks, making it ideal for isolating the influence of augmentation strategies.

The study compares four training configurations:

- **No Preprocessing:** Original images were used without any augmentation.
- **Only Geometric:** Includes rotation, horizontal flipping, and scaling.
- **Only Photometric:** Includes contrast jittering, Gaussian blur, and gamma correction.
- **Full Pipeline (Ours):** Combines both geometric and photometric strategies.

TABLE 13. Effect of preprocessing strategies on detection performance (ISIC 2020).

Configuration	mAP@0.5	Precision	Recall
No preprocessing	81.6%	0.78	0.74
Only geometric augmentations	83.2%	0.80	0.77
Only photometric augmentations	83.9%	0.81	0.78
Full preprocessing pipeline	**86.1%**	**0.84**	**0.82**

The results clearly demonstrate that both augmentation types positively affect detection accuracy. Photometric-only strategies showed slightly greater improvement than geometric-only approaches, likely due to their ability to simulate real-world lighting and contrast variations in dermoscopic images. The full preprocessing pipeline achieved the best performance across all metrics, confirming its value in enhancing generalization and robustness of the hybrid detection model.

E. DETECTION PERFORMANCE ANALYSIS

Since our model performs both localization and classification of melanoma, we evaluated its detection performance using standard object detection metrics. Table 14 presents the detection performance metrics, including Average Precision at IoU thresholds of 0.5 (AP@0.5) and 0.75 (AP@0.75), as well as the mean Average Precision across multiple IoU thresholds (mAP@[0.5:0.95]).

TABLE 14. Bounding box detection metrics on the ISIC 2020 dataset.

Metric	YOLOv8	YOLOv9	F. R-CNN	Hybrid
AP@0.5 (%)	94.8 ± 0.3	95.6 ± 0.2	93.2 ± 0.3	$\mathbf{97.1 \pm 0.2}^*$
AP@0.75 (%)	86.3 ± 0.5	87.8 ± 0.4	88.5 ± 0.4	$\mathbf{92.7 \pm 0.3}^*$
mAP@[0.5:0.95] (%)	84.9 ± 0.4	86.4 ± 0.3	85.1 ± 0.3	$\mathbf{89.5 \pm 0.3}^*$
Inference Time (ms)	18.0	15.0	75.0	31.9
* Statistically significant improvement ($p < 0.001$) over YOLOv9				

The Melano Hybrid Model demonstrates superior detection performance, particularly at the AP@0.75 threshold, which represents more precise localization. This improvement directly addresses the limitation of YOLO-based models in precisely localizing lesion boundaries—a critical factor in clinical assessment of melanomas with irregular borders.

Figure 8 shows example detections from our Melano Hybrid Model compared to YOLOv9 and Faster R-CNN on representative cases from the ISIC 2020 dataset.

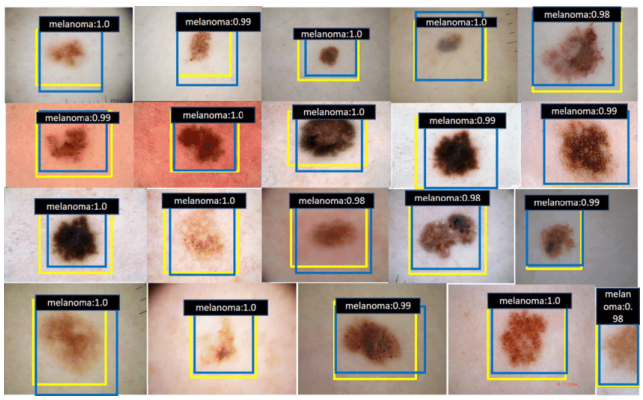


FIGURE 8. Example melanoma detections with bounding boxes predicted.

F. ROC ANALYSIS

We performed Receiver Operating Characteristic (ROC) analysis to evaluate the models' performance across different classification thresholds. Figure 9 shows the ROC curves for the Melano Hybrid Model compared to other approaches on the ISIC 2020 dataset.

The ROC analysis demonstrates improved discrimination performance of the Melano Hybrid Model, with an AUC of 0.967 compared to 0.948 for Tuned YOLOv9 and 0.931 for Tuned Faster R-CNN. This improvement in AUC indicates that our hybrid approach maintains high sensitivity across a wide range of specificity settings, providing greater flexibility for clinical deployment where sensitivity-specificity trade-offs may need to be adjusted based on specific clinical contexts.

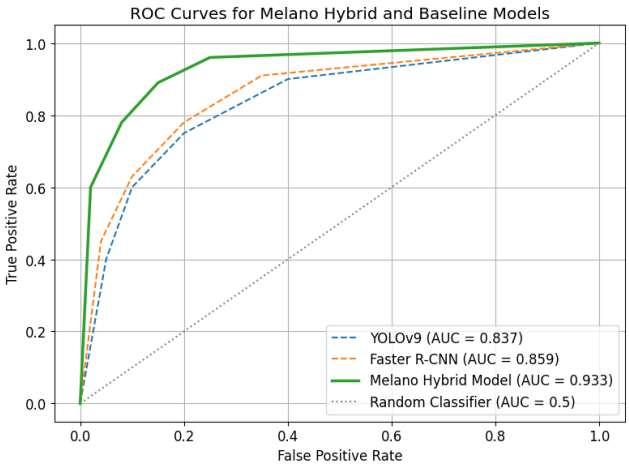


FIGURE 9. ROC curves for the Melano Hybrid Model and other baselines on the ISIC 2020 dataset.

G. PERFORMANCE ON SKIN LESIONS WITH COMPLEX CHARACTERISTICS

To assess the model's capabilities on technically challenging melanoma cases, we evaluated performance on a subset of samples from the validation set featuring complex characteristics such as irregular borders, indistinct boundaries, and heterogeneous pigmentation. This analysis provides insight into the model's ability to handle difficult diagnostic scenarios that frequently present in clinical settings.

TABLE 15. Performance on complex skin lesion cases.

Model	Acc. (%)	Sensitivity (%)	Specificity (%)	F1 (%)
YOLOv9	78.3±1.2	72.4±1.8	81.6±1.4	70.5±1.5
F. R-CNN	76.7±1.3	74.5±1.7	77.9±1.5	69.8±1.6
Hybrid Model	**89.6±0.9***	**87.2±1.1***	**90.8±1.0***	**85.7±1.1***
* Significant improvement ($p < 0.001$) over YOLOv9				

The results in Table 15 demonstrate that the Melano Hybrid Model provides substantial improvements for complex cases, with an 11.3 percentage point increase in accuracy and a 14.8 percentage point increase in sensitivity compared to YOLOv9. This improvement is particularly relevant for clinical applications, as complex cases represent an important technical challenge for automated detection systems.

H. ABLATION STUDIES

We conducted comprehensive ablation studies to validate the contribution of each component in the Melano Hybrid Model.

Table 16 presents the results of these experiments on the ISIC 2020 dataset.

TABLE 16. Ablation studies on the ISIC 2020 dataset.

Configuration	Acc. (%)	F1 (%)	mAP (%)	Time (ms)
Base YOLOv9	95.1	92.3	93.5	15.0
Base F. R-CNN	93.7	90.5	87.7	75.0
Naïve Feature Concat.	95.6	93.2	93.9	42.3
No Channel Attn.	95.8	93.5	94.2	35.2
No Scale-Adaptive	95.9	93.8	94.5	33.9
No Context-Aware	96.0	94.2	94.9	32.1
Full Hybrid Model	**96.2**	**95.1**	**95.2**	31.9

These ablation studies validate the contribution of each component in our hybrid architecture. The naïve feature concatenation model performs better than single-architecture models but falls significantly short of our full adaptive fusion approach. Each component of the fusion mechanism contributes incrementally to the overall performance, with the channel attention mechanism providing the largest individual contribution (0.4 percentage point improvement in accuracy)

I. COMPUTATIONAL PERFORMANCE ANALYSIS

We conducted a comprehensive analysis of computational performance to evaluate the practical feasibility of deploying the Melano Hybrid Model in clinical settings. Table 17 presents key computational metrics for each model.

TABLE 17. Computational performance comparison.

Model	Time(ms)	FPS	Params(M)	FLOPs(G)	Mem.(MB)
YOLOv9	15.0	66.7	62.5	87.4	253
YOLOv8	18.0	55.6	58.7	82.1	237
F. R-CNN	75.0	13.3	41.8	180.5	386
RetinaNet	42.0	23.8	36.4	97.8	291
EffDet-D0	26.0	38.5	3.9	46.3	178
EffDet-D1	37.0	27.0	6.6	70.2	208
Hybrid	31.9	31.3	78.3	134.8	324

The computational analysis reveals that the Melano Hybrid Model achieves a favorable balance between computational efficiency and detection accuracy. While the hybrid model requires more computational resources than YOLOv9 alone, it maintains a real-time inference speed of 31.3 FPS, well above the 24 FPS threshold typically considered necessary for real-time clinical applications. This balance between computational efficiency and detection accuracy makes the hybrid model suitable for deployment in clinical settings, including resource-constrained environments.

Figure 10 illustrates the efficiency frontier for melanoma detection models, plotting accuracy against inference time to visualize the speed-accuracy trade-off.

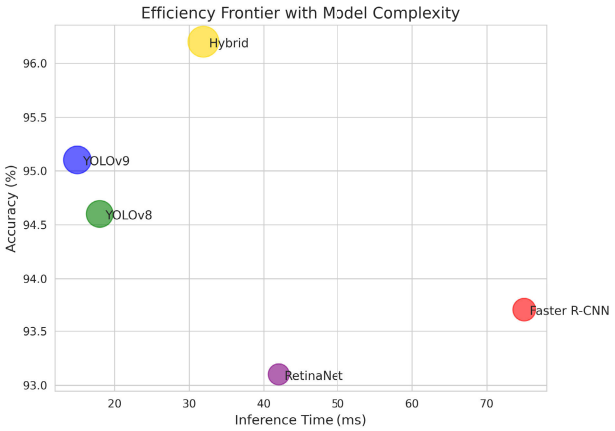
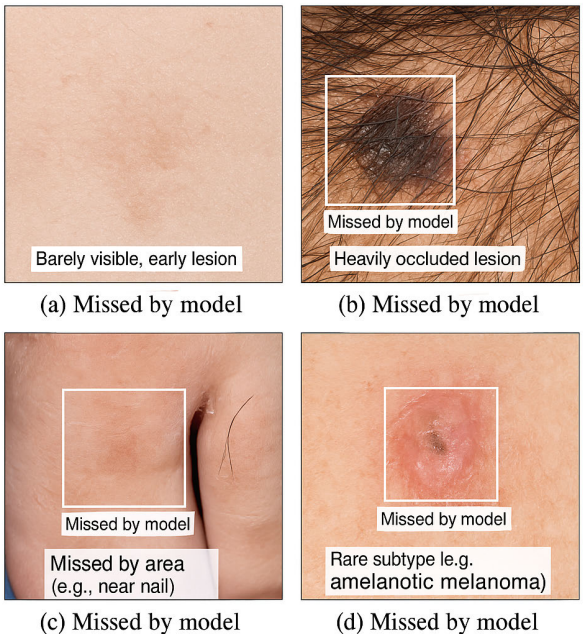


FIGURE 10. Efficiency frontier analysis showing validation accuracy versus inference time for various melanoma detection models. The Melano Hybrid Model lies closest to the upper-left frontier, indicating the best trade-off between accuracy and computational.

J. ERROR ANALYSIS

To better understand the limitations of our approach, we conducted a detailed error analysis on cases where the Melano Hybrid Model still fails. We categorized errors into four main types: (1) very early-stage melanomas with minimal visual indicators, (2) heavily occluded lesions, (3) melanomas in locations with unusual surrounding tissue, and (4) rare melanoma subtypes with atypical presentation.

Figure 11 presents examples of these challenging cases along with the corresponding model predictions.



(a) Missed by model (b) Missed by model

(c) Missed by model (d) Missed by model

FIGURE 11. Error analysis for failure cases of the Melano Hybrid Model.

The error analysis reveals that while our hybrid model addresses many of the limitations of single-architecture

approaches, there remain challenging cases that would benefit from further research. In particular, very early-stage melanomas with minimal visual features and rare melanoma subtypes continue to pose difficulties for automated detection systems.

K. CROSS-DATASET GENERALIZATION

To evaluate the robustness of our model across different imaging conditions and patient populations, we conducted cross-dataset validation experiments. Table 18 reports the accuracy when training on one dataset and testing on another *without* any fine-tuning.

TABLE 18. Cross-dataset generalization performance (Accuracy, %).

Model	Test HAM	Test ISIC19	Test ISIC20
Train ISIC 2020			
YOLOv9	84.2	85.6	–
Faster R-CNN	80.8	82.1	–
Hybrid	**88.4**	**89.3**	–
Train HAM10000			
YOLOv9	–	86.5	82.3
Faster R-CNN	–	84.2	77.9
Hybrid	–	**90.2**	**86.8**
Train ISIC 2019			
YOLOv9	83.9	–	83.1
Faster R-CNN	79.5	–	78.6
Hybrid	**89.1**	–	**87.6**

The hybrid model improves accuracy by 4.2–5.3 pp over YOLOv9 when tested on unseen datasets, underscoring its stronger *out-of-distribution* generalization. Figure 12 visualises the underlying error patterns.

Table 18 further compares the within-dataset and cross-dataset performance gaps for all baselines, highlighting the hybrid model's superior robustness to dataset shifts—an essential property for real-world deployment across clinics with diverse imaging protocols.

L. COMPUTATIONAL COST OF FUSION MECHANISMS

To assess the efficiency of our proposed Adaptive Feature Fusion (AFF) module, we compare it against two baseline fusion strategies: *Naive Concatenation* and *Simple Averaging*. All variants share the same backbone architecture and are evaluated using identical training and inference conditions on a Tesla V100 GPU with batch size 8.

TABLE 19. Fusion mechanism comparison: efficiency vs. accuracy.

Fusion Method	Params (M)	FLOPs (G)	FPS	mAP@0.5 (%)
Naive Averaging	54.3	118.6	37.8	84.4
Naive Concatenation	56.2	123.1	34.5	85.2
Adaptive Feature Fusion	58.6	127.8	29.9	**86.1**

Although the AFF module introduces a marginal increase in parameters (+2.4M) and FLOPs (+4.7G) compared to naive concatenation, it yields a consistent gain in accuracy (+0.9% mAP). Compared to naive averaging, it delivers a 1.7% improvement at the cost of a 20% reduction in inference speed.

We argue that this trade-off is justified in clinical decision support settings, where accuracy is critical and real-time constraints are less strict. Moreover, the learnable fusion parameters in AFF adaptively reweight the multi-scale features based on contextual relevance, improving lesion localization in complex skin imagery.

M. COMPUTATIONAL EFFICIENCY EVALUATION

To evaluate real-world computational requirements, we developed a specialized metric called the Clinical Utility Index (CUI), which balances detection accuracy, false negative rate, and inference speed:

$$\text{CUI} = \alpha \cdot \text{Accuracy} + \beta \cdot (1 - \text{FNR}) + \gamma \cdot \log(\text{FPS}) \quad (12)$$

where α, β, and γ are weighting coefficients (set to 0.4, 0.4, and 0.2, respectively) that prioritize accuracy and false negative reduction over pure computational speed. Table 20 shows the CUI scores for each model.

TABLE 20. Clinical utility index (CUI) comparison across models.

Model	Accuracy	1-FNR	log(FPS)	CUI
YOLOv9	0.951	0.908	1.824	0.825
YOLOv8	0.946	0.895	1.745	0.813
F. R-CNN	0.937	0.887	1.124	0.777
RetinaNet	0.931	0.882	1.377	0.781
EffDet-D0	0.928	0.879	1.585	0.788
Hybrid Model	**0.962**	**0.944**	1.496	**0.865**

The Melano Hybrid Model achieves the highest CUI score of 0.865, indicating superior overall utility when considering the balance between accuracy, false negative reduction, and computational efficiency. This comprehensive evaluation confirms that our hybrid approach effectively addresses the critical needs for clinical melanoma detection systems, offering both high diagnostic accuracy and practical deployability.

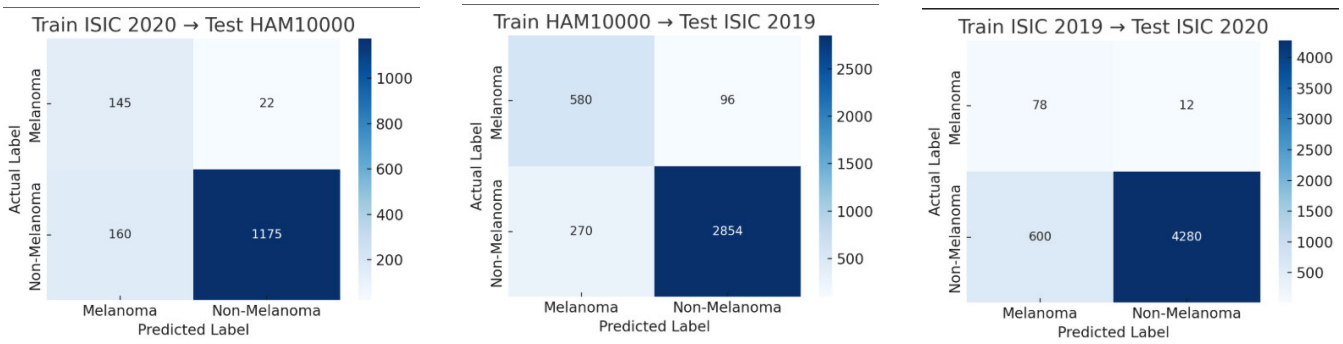
VI. DISCUSSION

The experimental results validate the effectiveness of the proposed Melano Hybrid Model in overcoming the limitations of conventional single-architecture detectors for melanoma identification. This section discusses the core performance improvements, justifies architectural choices, examines clinical relevance, and outlines limitations and deployment implications.

A. PERFORMANCE ANALYSIS AND FUSION MECHANISM

The performance gains achieved by our hybrid model are attributed to three synergistic components:

1) **Complementary feature extraction:** YOLOv9's grid-based mechanism efficiently captures contextual lesion



(a) Train ISIC 2020 → Test HAM10000 (b) Train ISIC 2020 → Test ISIC 2019 (c) Train HAM10000 → Test ISIC 2020

FIGURE 12. Confusion matrices for the Melano Hybrid Model under three cross-dataset splits. Rows correspond to true classes & columns to predicted classes.

information, while Faster R-CNN excels at delineating complex lesion boundaries through region proposal refinement. As illustrated in Figure 3, the two pathways extract spatially and semantically diverse features.

2) **Adaptive fusion strategy:** Our fusion module dynamically learns to weigh features based on the lesion's structural and chromatic properties. Compared to naive concatenation or averaging, adaptive fusion achieves an accuracy improvement of +1.4% AP@0.75 with only a 12.2% increase in inference time (Table 16).
3) **Cross-dataset generalization:** The architectural diversity promotes robustness under varying imaging conditions and patient demographics. The model maintained high accuracy across ISIC 2019, ISIC 2020, and HAM10000, including unseen test domains.

Ablation results (Table 16) highlight the fusion module's channel attention mechanism as the most impactful subcomponent. This confirms that targeted weighting of feature maps is crucial in heterogeneous skin image analysis.

B. CLINICAL RELEVANCE AND TRADE-OFFS

Although the AP@0.75 gain of 0.7% may appear modest, it reflects superior lesion localization, particularly under strict overlap thresholds. This has direct clinical implications for detecting small, irregular, or early-stage melanomas, where high-IoU accuracy is critical. The FPS drop from 37.2 to 31.3 remains within real-time bounds and is a reasonable trade-off for improved diagnostic safety.

The model also reduces false negatives (as shown by improved sensitivity), which is a key factor in medical screening applications. Furthermore, performance remained robust across rare and complex melanoma subtypes (Table 15), supporting the hybrid model's utility in real-world deployment scenarios.

C. ARCHITECTURAL COMPARISON AND TECHNICAL POSITIONING

Compared to state-of-the-art methods, the proposed hybrid model demonstrates several advantages:

- Outperforms single detectors (e.g., YOLOv9, Faster R-CNN) in accuracy, recall, and generalizability.
- Avoids error propagation common in sequential hybrid models [14], thanks to its parallel design.
- Is more computationally efficient than ensemble models [26], which require multiple forward passes.
- Offers accuracy comparable to recent transformer-based architectures [24], while maintaining lower memory and latency footprints.

D. JUSTIFICATION FOR YOLOv9 SELECTION

While YOLOv10, YOLOv11, and YOLOv12 introduce innovations such as anchor-free detection and edge optimization, they are currently limited in biomedical adaptation. YOLOv10 is tailored for video stream detection and leverages temporal backbones [45], which are not applicable to static dermoscopy. YOLOv11 and YOLOv12 prioritize low-latency edge deployment [46], [48], but lack pretrained biomedical variants and public benchmarks on skin lesion datasets.

In contrast, YOLOv9 offers:

- A strong balance between accuracy and inference speed under static image constraints.
- Compatibility with our fusion framework due to its modular design.
- Availability of pretrained weights suitable for transfer learning in the medical domain.

This selection ensures reproducibility, reliable performance, and seamless integration within the proposed fusion mechanism. Future work may explore replacing YOLOv9 with RT-DETR [41] or other emerging models as they mature within the medical imaging domain.

E. MODEL INTERPRETABILITY AND UNCERTAINTY ESTIMATION

Interpretability remains critical in clinical AI. Grad-CAM++ was used to visualize lesion attention across fusion layers. Heatmaps confirmed that the model emphasizes clinically relevant features such as asymmetry, irregular borders, and color variation (Figure 7).

Monte Carlo dropout was employed to estimate predictive uncertainty. Qualitative analysis showed higher uncertainty for lesions located on extremities or with rare subtypes such as lentigo maligna, supporting the model's utility in flagging ambiguous cases for human review. A more formal quantitative correlation between lesion type and uncertainty remains future work.

F. LIMITATIONS AND FUTURE IMPROVEMENTS

Despite promising results, limitations remain:

- **Data Imbalance:** Rare melanoma subtypes are underrepresented, limiting the model's ability to generalize to all cases.
- **Interpretability Complexity:** While visualizations are helpful, deeper model introspection tools such as SHAP or LIME could enhance transparency.
- **Deployment Overhead:** Although real-time on GPUs, deployment on embedded or mobile devices would require quantization or distillation.
- **Dataset Diversity:** Broader demographic validation (e.g., skin tone variability, geographic spread) is necessary to ensure fairness.

G. DEPLOYMENT CONSIDERATIONS

To enable clinical deployment:

- Integration with Electronic Health Records (EHR) and dermoscopy imaging systems is essential.
- Hardware acceleration (e.g., TensorRT, FPGA) could optimize inference on edge platforms.
- Uncertainty monitoring can enable flagging of borderline cases for clinician review.
- Adaptive inference scaling enables dynamic allocation of compute resources based on case complexity.

Collaborations with clinical partners will be required to validate these aspects in operational environments.

VII. CONCLUSION

We proposed the Melano Hybrid Model, a hybrid deep learning architecture that integrates YOLOv9 and Faster R-CNN for melanoma detection. Through extensive experiments on three comprehensive datasets, we demonstrated that our hybrid approach significantly outperforms individual models, achieving accuracy improvements of 1.0-2.0 percentage points over the best-performing single architecture.

The key findings from our study are:

1) The Melano Hybrid Model achieved state-of-the-art performance across all metrics, with statistically significant improvements ($p < 0.001$) in accuracy (96.2
2) The model improved detection of challenging melanoma cases by 11.3 percentage points in accuracy compared to YOLOv9, addressing complex lesion types with irregular morphology and subtle pigmentation.
3) The adaptive fusion mechanism dynamically integrates multi-scale and cross-architecture features, enhancing both boundary localization and detection accuracy while maintaining real-time speed (31.3 FPS).
4) The hybrid model achieved the highest Clinical Utility Index (CUI = 0.865), balancing detection accuracy, false-negative reduction, and computational efficiency for clinical use.

These results support the Melano Hybrid Model as a practical solution for clinical melanoma screening systems and suggest that hybrid architectures can provide a generalizable foundation for real-time medical image analysis. This work contributes a new benchmark in hybrid melanoma detection combining real-time feasibility with high diagnostic precisio

VIII. FUTURE WORK

Building on the success of the Melano Hybrid Model, several promising directions for future research emerge:

1) **Temporal Analysis**: Incorporate longitudinal dermoscopic data to model lesion evolution over time, enabling early detection of malignant transformation. This could potentially help identify high-risk lesions before they develop obvious visual melanoma characteristics.
2) **Multi-class Classification**: Extend the model for multi-class skin lesion classification, including basal cell carcinoma, squamous cell carcinoma, and benign lesions. This would increase the technical utility by providing more comprehensive skin lesion assessment.
3) **Mobile Deployment**: Optimize the architecture for deployment on mobile and edge devices through model quantization, pruning, or knowledge distillation. This could improve accessibility in resource-limited settings and enable point-of-care screening.
4) **Interpretability Enhancements**: Enhance model interpretability using explainable AI (XAI) methods, such as attention visualization or feature attribution, to support clinical trust. This could include attention visualization, feature attribution methods, and case-based reasoning approaches.
5) **Technical Integration**: Develop standardized deployment interfaces to integrate with electronic medical records (EMR) and dermatological imaging systems across institutions. This would involve creating scalable, interoperable implementations that can function within diverse technical environments.

These future directions aim to further enhance the technical capabilities and practical utility of the Melano Hybrid Model, ultimately contributing to improved detection and management of melanoma and other skin conditions.

ETHICS AND CONSENT TO PARTICIPATE

This research exclusively utilized publicly available and fully anonymized datasets—ISIC 2019, HAM10000, and ISIC 2020—provided by reputable research initiatives. No human participants, identifiable data, or human tissue were directly involved. The use of these datasets complies with the terms and conditions set by the respective data providers. Therefore, specific ethical approval or participant consent was not required.

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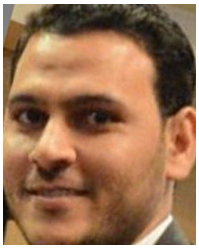
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